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1.1.1. Introduction

1.1.1.1. Project Overview

The Padma Multipurpose Bridge is a two-level, road-rail steel truss (*A truss bridge is a bridge that uses a rigid framework of interconnected triangles to support heavy loads over large spans*) bridge spanning the Padma River which is one of the widest and most hydrologically active rivers in South Asia (Bangladesh Bridge Authority, n.d.) . At 6.15 kilometers long and 18.1 to 21.1 meters wide, it stands as Bangladesh's largest bridge and the first fixed road crossing built over the Padma. The structure features a four-lane highway on its upper deck and a single-track, dual-gauge railway on its lower deck. Beyond transport, it functions as a critical multi-utility corridor carrying a high-voltage electricity transmission line, a natural gas pipeline, and fiber-optic cables.



From Wikimedia Commons



Constructed under the Bangladesh Bridge Authority (BBA) which is an arm of the Bridges Division within the Ministry of Road Transport and Bridges. The bridge connects Louhajong upazila in Munshiganj district at the Mawa end on the north bank with Shariatpur and Madaripur districts at the Janjira end on the south bank. Officially

inaugurated on June 25, 2022, and opened to traffic the following day, this major engineering effort directly links Dhaka with 21 historically isolated southwestern districts, integrating the region into the national road and rail grid.

1.1.1.2. Context

Before the Padma Bridge was built, the only way to cross the Padma River was by ferry or speedboat. These trips were slow and unpredictable, taking anywhere from a few hours to an entire day. Bad weather, heavy fog, and strong river currents frequently brought travel to a complete stop. This created a massive bottleneck that isolated 30 to 35 million people across 21 southwestern districts, hurting local trade, industrial growth, and access to healthcare in a region with above-average poverty levels.

The idea for a permanent river crossing started in 1998 after the completion of the Bangabandhu Bridge. In 2004, a study by JICA(Japan International Cooperation Agency) confirmed the route between Mawa and Janjira, and the project officially gained government approval in August 2007 with an initial budget of Tk 101.61 billion for a road bridge. The design was later expanded to include a railway deck, with detailed engineering completed between 2009 and 2011 by Maunsell-AECOM (funded by the ADB(Asian Development Bank)) and reviewed by the UK firm Flint & Neill.

Building the bridge was a huge technical challenge because the Padma is a wide, powerful river with riverbed erosion reaching depths of up to 62 meters.

The project also faced a major funding crisis in 2012 when the World Bank canceled its \$1.2 billion loan over corruption allegations, leading other international lenders like the ADB, JICA, and Islamic Development Bank to withdraw as well. Instead of cancelling the project, the Bangladesh government decided in January 2013 to fund the entire megaproject from its own national budget, turning the bridge into a lasting symbol of national self-reliance.

1.1.1.3. Key Stakeholders

Stakeholder	Stakeholder's role	Category	Type
Bangladesh Bridge Authority (BBA)	Project sponsor, overall executing agency, and facility owner under the Ministry of Road Transport and Bridges.	Project Owner & Client	Direct
Executive Committee of the National Economic Council (ECNEC)	Government entity that approved budget revisions and authorized full national funding.	Regulatory & Approval Body	Direct
AECOM Consortium	Lead consultant (led by Dr. Robin Sham, alongside SMEC, Northwest Hydraulic, ACE, Aas-Jakobsen, and HR Wallingford) for detailed design.	Design Consultant	Direct
Flint & Neill, UK (now COWI)	Provided independent verification and checks on detailed bridge engineering.	Independent Design Checker	Direct
China Major Bridge Engineering Company Ltd. (CMBEC)	Primary civil works contractor for the construction of the main bridge superstructure and foundations.	Main Bridge Contractor	Direct
Sinohydro Corporation (China)	Contractor responsible for bank protection and river channel stabilization works.	River Training Contractor	Direct
Korea Expressway Corporation Consortium	Supervision consultant (with PEC, KECC, Sunjin, ACE, and DevConsultants) overseeing construction quality and execution.	Main Bridge Supervision	Direct
Bangladesh Army (SWO-West) with BRTC & BUET	Supervision team for approach roads, toll plazas, and service areas.	Infrastructure Supervision	Direct
Korea Expressway Corporation & CMBEC	Joint operators appointed post-completion for toll collection and routine facility maintenance.	Operations & Maintenance	Direct
Local Landowners (Munshiganj, Shariatpur, Madaripur)	Directly affected households (~917 hectares acquired) receiving resettlement and land compensation.	Affected Local Communities	Direct

Stakeholder	Stakeholder's role	Category	Type
11-Member Panel of Experts	Technical oversight group chaired by Prof. Jamilur Reza Chowdhury providing expert advice to BBA.	Independent Technical Advisory	Indirect
World Bank (IDA), ADB, JICA, & IsDB	Original co-financiers whose initial pledges ended after the World Bank's 2012 withdrawal.	Former Intended Lenders	Indirect
Anti-Corruption Commission (ACC)	Independent national body that investigated procurement allegations associated with the project.	Investigative Agency	Indirect
Regional Residents & National Economy	Over 30 million residents across 21 southwestern districts, freight operators, and businesses benefiting from connectivity.	Public End Users	Indirect

1.1.2. Objectives and Scope

1.1.2.1. Project Objectives

Objective	Description	Target outcome
1. All weather River crossing	Build a 6.15 km long two level road and rail bridge across the Padma River.	Replace slow, risky and weather dependent ferry services
2. Direct Regional Connection	Connect the capital city Dhaka directly to 21 south-western districts.	Provide direct road and rail links for roughly 30 million people in an isolated region.
3. Trade and Port Linkage	Link with national road and rail network with Mongla Port, Payra Port and other routes.	Create a continuous transportation for international trade
4. Save Travel Time and Cost	Reduce the transit duration and fuel expenses for passenger	Cut river crossing duration from several hours down to about 10 minutes.
5. Shared Utility Corridor	Place high voltage power lines, gas pipelines and fiber optic internet cables inside the bridge.	Build a single bridge structure that carries transport and essential utilities together.
6. Support Economic and Social Welfare	Help local businesses expand, attract new industrial units and generate more employment in rural districts.	Add 1.0 to 1.23% to national GDP and expand the south-west regional economy by roughly 2.3% annually.
7. Local Engineering Skills	Collaborate with international experts to provide local teams practical experience during implementation.	It builds real world skills for Bangladeshi engineers.

1.1.2.2. Scope of the Project

Category	Simple Description	Key deliveries	Scope Status
1. Main Bridge Structure	Build the 6.15 km two-level steel bridge spanning across 41 river spans.	In upper side 4-lane highway and lower side single-track dual-gauge rail line connecting Mawa and Janjira	In-Scope
2. Deep Foundation Works	Drive extra-deep foundation piles into the riverbed so the bridge piers stay safe and stable.	Installation of 262 steel support pipes under the river and 365 concrete foundation posts for the connecting approach bridges.	In-Scope
3. River Training Works	Protect the riverbanks and hold the river channel in place so water currents do not wash away land.	About 14–15 km of bank protection, and geo-bag placement on both riverbanks .	In-Scope
4. Approach Infrastructure	Construct connecting highways, toll booths and support facilities on both sides of the river.	About 10–13 km of approach roads, two service areas, toll plazas and site offices.	In-Scope
5. Overland Rail Link Network	The 170–215 km long railway line extension connects Dhaka to Jashore beyond the main bridge.	Its like a separate fast-track project by Bangladesh Railway with its own budget	Out-of-Scope
6. Daily Tolls and Future Maintenance	Collecting tolls daily and repairing the road surface after the bridge opened	Handled under a separate operational phase by external operating teams appointed after completion	Out-of-Scope
7. Regional Water Projects	Nearby river management and irrigation schemes, such as the proposed Padma Barrage.	Planned and executed as independent civil projects totally outside the main bridge budget.	Out-of-Scope

1.1.3. Project Management Processes

1.1.3.1. Methodology

The Padma Multipurpose Bridge Project was following a Design Bid Build (DBB) model, where the detailed design was provided by a professional engineering service company, AECOM. The design was certified by Flint & Neill UK. After the finalization of the design, each segment of the project was divided into contracts and then subcontracted foreign companies were hired. China Major Bridge Engineering was contracted for the construction of the Padma multi-purpose bridge. The DORP NGO helped implement the resettlement activities under the supervision of CSC Cell, Bangladesh Army and the Corps of Engineers (Railway Technology, n.d.).

This approach contributed towards minimizing the overall cost and schedule changes at each stage of the project, which were formally routed through ECNEC (The Daily Star, 2023). This gave the project a series of clearly documented baselines, which helped manage the interdependent contracts with a strong centralized programme.

1.1.3.2. Planning and Design

1.1.3.2.1. Planned Schedule

After the Japan International Cooperation Agency (JICA) finished their feasibility study in 2005, it was shown that this project would help the regional growth immensely as it'd bring reduced regional poverty, improved public services and trade routes et cetera (Rahman & Kabir, 2022). Even though Prime Minister Sheikh Hasina laid the symbolic foundation stone back in 2001, the original plan to start working on the construction of the building was set to 2011 and to finish by 2013 (The Daily Star, 2022).

1.1.3.2.2. Estimated Cost

Milestone	Estimated Cost	Date
Original concept approval (Only road)	Tk.101.61 billion	August 2007
Design phase (Phase 1)	N/A	29 January, 2009
Revised design approval (Railway Component added)	Tk.205.07 billion	January, 2011
World Bank financial agreement signed	US\$1.2 billion	April 2011
Main Bridge construction contract	Tk.12,133 crore	17 June, 2014
Fully self-financed after donor withdrawal	Tk.30,193.7 million	2013-2022
VAT/tax increases on contractor payments; completion extended to June 2024	Tk 32,605 crore	2022–2023

1.1.3.3. Execution

1.1.3.3.1. Schedule

After deciding not to rely on loans from funding agencies, the government of Bangladesh decided to complete the project from the GoB Fund in 2012 (Padma Multipurpose Bridge Project, n.d.), handing the construction contract to China Major Bridge Engineering Co. Ltd. to build the main portion of the bridge. The project was rescheduled a few times, which made starting the construction of the bridge itself quite difficult, starting in November of 2014 (The Daily Star, 2022) .

Because of the huge allegation of corruption in late 2011, fundings were cancelled and co-financiers such as JICA, ADB either withdrew or stalled their financing (Al Jazeera, 2012; MP-IDSA, 2012). After starting the initiation of self financing, the timeline was delayed by nearly three years as they had restructured the financing model and issued a new international tender process to secure new contractors (Global Anti-Corruption Blog, 2015; Wikipedia, 2022).

The first span was finished in 2017, placing the first 3000 ton steel span which made the structure physically visible above water. While the foundation of the bridge was completed by placing all the spans within 2020, the completed work with roadway, power transmission lines and rail slabs weren't done before 2022 (The Daily Star, 2022; The Business Standard, 2020).

1.1.3.3.2. Cost

Project Component	Allocated Amount	% of Total Budget
Main Bridge Construction	₹12,133.39 crore	~40.2%
River Training Work	₹9,400.00 crore	~31.1%
Land Acquisition	₹2,698.73 crore	~8.9%
Approach Roads & Toll Plazas	₹1,907.68 crore	~6.3%
Resettlement	₹678.37 crore	~2.2%
Environment & Others	₹1,860.20 crore	~6.2%

1.1.4. Challenges and Solutions

1.1.4.1. Problems Encountered

Political Challenges:

i) Geopolitical tension and Financial withdrawal:

[According to BBC](#), (2012) due to suspicions of an alleged bribery, the World Bank cancelled its \$1.2 billion credit in June 2012. After this incident, many other financiers like ADB, JIC, IDB withdrew from the project which led to the stalling of the project.

ii) Discontinuation:

The political and financial fallout with the financiers led to a cascading effect of instability, as the project director was suspended, the bridges division secretary was transferred and at the peak of the scandal, the communication minister Syed Abul Hossain resigned ([Staff Correspondent, 2012](#)).

Economic Challenges:

i) Foreign Reserve Depletion and High Capital:

After the financial setback, it was decided by then prime minister Sheikh Hasina (Apa) to finance this project using Bangladesh's own reserve. Unfortunately, this had put a high strain on our national reserves, especially the foreign exchange reserves, due to the dependency on foreign contractors.

ii) Cost Escalations and Budget Adjustments:

Due to newer additions to the projects, the estimated cost kept rising and due to budget provisions, VATs and taxes were raised to accommodate the cost.

Social Challenges:

i) Land Acquisition and Human Displacement:

Many communities, businesses and people who owned land near the riverbanks were displaced after the acquisition of around [917 hectares](#) of land (Hossain, 2023).

ii) Human Safety amidst the Pandemic:

During the COVID-19 Lockdowns, the government faced difficulties to ensure proper safety of the thousands of workers on the project all while trying to comply with the public health mandates.

Technological Challenges:

i) Complexities in Pier foundation designs:

Conventional pier designs were ineffective due to softer soils underneath and the heavy loads, which led to lengthy pauses and redesigns. [14 piers](#) were specifically found to have design flaws (Sultana, 2017).

Environmental Challenges:

i) Scour Issues and unpredictable waters:

The 6 km long bridge faced extreme currents from the Padma river. The bed was also shifted and sediments were piled up creating heavy loads. The current also created scour holes which compromised the structural integrity.

ii) Natural Hazards and Erosions:

Close proximity towards active himalayan plates created risks of seismic events. Also, rapid riverbank erosion was endangering project sites and key infrastructures.

Legal Challenges:

i) Compliance and Anti-Corruption Governance:

Integrity allegations required extensive internal reviews, contractual restructuring, and stricter governance mechanisms to maintain public credibility and satisfy domestic regulatory standards.

ii) Contractual Liability and International Protocols: Navigating complex international procurement contracts across multiple jurisdictions introduced legal friction regarding schedule slippage, scope variations, tax law changes, and contractor claims.

1.1.4.2. Solutions Implemented

Political Solutions:

i) Sovereign Financial Pivoting:

Following the withdrawal of international financiers, the Government of Bangladesh made a strategic decision to [self-fund](#) the entire \$3.87 billion project using national reserves, eliminating external geopolitical dependency and restoring project momentum (Bhaumik, 2013).

ii) Institutional Restructuring and Military Oversight:

To counteract leadership instability and maintain project governance, the Bangladesh Army's Construction Supervision Consultant (CSC) unit and the 99 Composite Brigade were deployed to provide administrative stability, field management, and overall site security.

Economic Solutions:

i) Foreign Reserve Management and Staged Disbursements:

To cushion the impact on national reserves and handle foreign exchange demands, Bangladesh's central bank and treasury executed phased dollar disbursements tied directly to project milestones completed by foreign contractors.

ii) Fiscal Reallocation and Revised Project Budgets:

The government approved successive revisions to the Development Project Proposal (DPP/RDPP), adjusting budget allocations to absorb currency depreciation, import duties, expanded river training works, and scope additions like the rail-link integration.

Social Solutions:

i) Comprehensive Resettlement Villages and Rehabilitation:

To address human displacement across 917 hectares, the Bangladesh Bridge Authority constructed modern resettlement centers equipped with utilities, schools, and medical facilities, alongside providing cash compensation above market rates and livelihood training (Hossain, 2023).

ii) Managed Bio-Bubble System:

To safeguard workforce health during COVID-19, the site was converted into a strict, closed ["bio-bubble"](#) where workers lived in isolated quarters with access to medical care, allowing construction to continue safely during nationwide lockdowns (Habib, 2022).

Technological Solutions:

i) Advanced Skin Grouting and Heavy Pile Redesign:

To resolve the structural issues on the 14 problematic piers, engineers expanded the pile foundation design from six to eight driven steel piles and utilized high-pressure chemical "skin grouting" to artificially solidify the surrounding alluvial sand layers (Sultana, 2017).

ii) Deployment of High-Capacity Machinery:

To overcome technical installation challenges, specialized heavy equipment including custom 3,500 kJ hydraulic hammers and a 3,600-tonne floating crane was deployed to drive 3-meter-diameter steel piles up to 122 meters deep into the riverbed.

Environmental Solutions:

i) Extensive River Training Works (RTW):

To combat extreme currents, bed shifts, and scour holes up to 62 meters deep, the project executed massive river training efforts using rock rip-rap, geotextile sheets, and submerged concrete block mattresses to protect the pier foundations and stabilize the banks.

ii) Seismic Isolation and Structural Safeguards:

To protect against Himalayan seismic risks and vessel collisions, the bridge was equipped with state-of-the-art Friction Pendulum Bearings (FPBs) to absorb heavy earthquake shocks, along with heavy-duty collision buffers built into the pier caps.

Legal Solutions:

i) Anti-Corruption Compliance Frameworks:

Internal investigations and Anti-Corruption Commission (ACC) inquiries were conducted alongside updated governance oversight protocols to satisfy domestic regulatory standards and restore public trust in project execution.

ii) Multi-Jurisdictional Contractual Alignment:

Legal and procurement teams established standardized operational guidelines and unified international dispute-resolution protocols across foreign entities (such as Chinese contractors, Korean consultants, and Western design firms) to resolve scope variations and tax compliance claims smoothly.

1.1.5. Results and Performance

1.1.5.1. Outcome

The Padma Multipurpose Bridge was inaugurated on 25th June, 2022 and opened to road traffic on 26th June, 2022. It became Bangladesh's largest bridge. Moreover, it improved the transportation system between Dhaka and the South-Western part of the country.

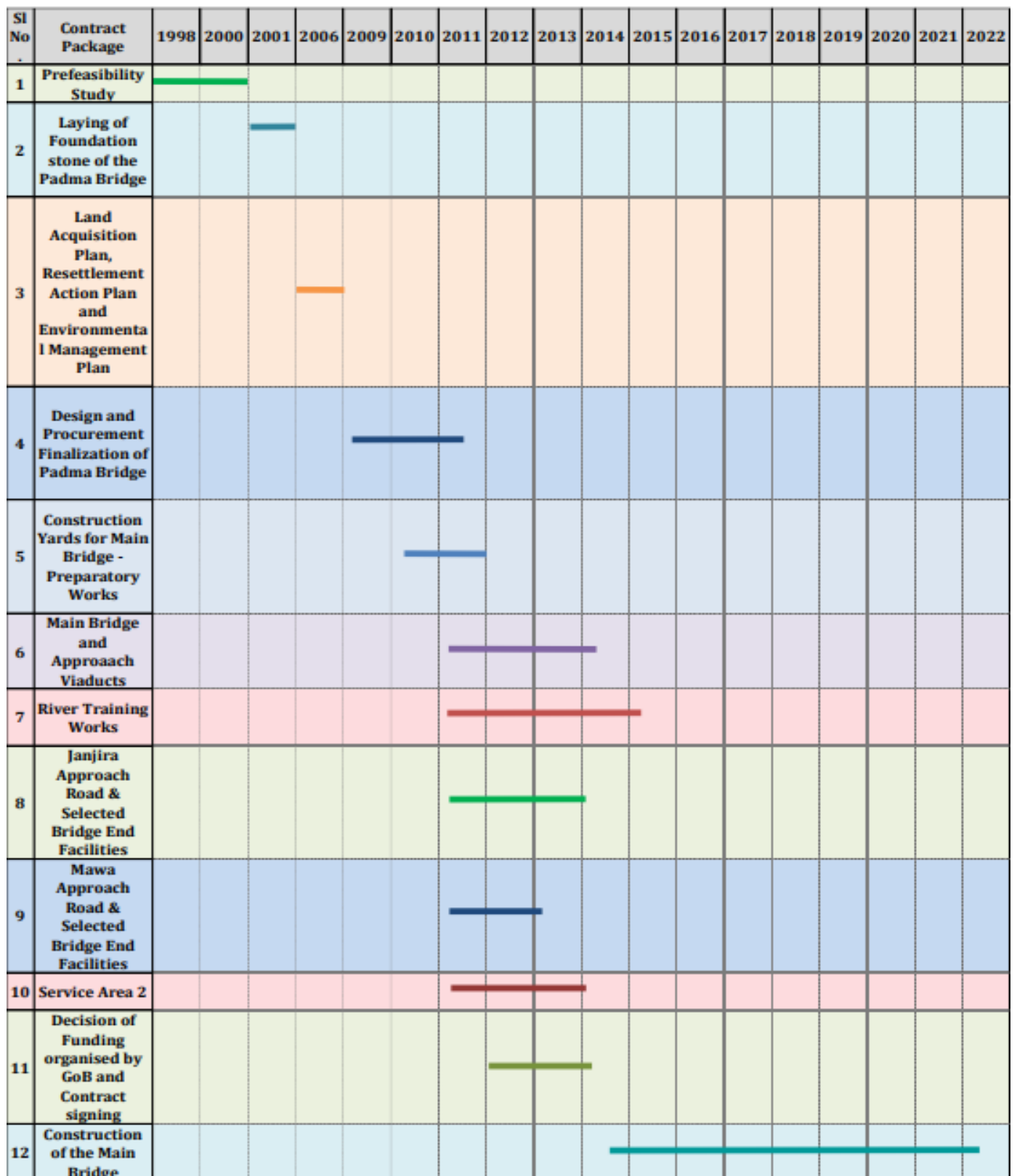
The main outcomes from the project:

1. It reduced travel time between Dhaka and the South-Western districts. Moreover, it benefits around 30 million people.
2. A journey took several hours by ferry in the past. But, it takes around 10 minutes to cross the bridge right now.
3. It directly connects Dhaka with a total of 21 South-Western districts. For this reason, it benefits around 30 million people.
4. Mainly it improves the connection to the Mongla and Payra seaports directly.
5. This bridge also carries a gas pipeline, power transmission line and fibre-optic cable. It makes the bridge more important than just a rail or road bridge.
6. Though this project went through many barriers like financial difficulties, difficult soil conditions, and the COVID-19 pandemic, the project was completed.
7. This project was completed with mostly domestic financing which proved that Bangladesh was able to complete a large infrastructure project by using its own resources.
8. Tk. 1835 crore saved from the estimated budget of this project via "Contractionary Policy at close-out"

1.1.5.2. Performance Metrics

Metric	Plane	Actual
Construction start	2011-2012	26th November, 2014 (main bridge pilling)
Opening	2013-2015	25-26 June 2022 (opened to traffic) Financial close-out 30 June, 2024
Project Cost	In 2007: Tk.10,161 crore In 2011: Tk.20,502 crore	Tk.301.93 billion (approximately US\$3.6-3.87 billion)
Overall Cost Variance	N/A	Approximately 3x the original 2007 estimate; Tk. 1835 crore saved versus the last approved 2022/23 estimate
Foundation Depth (engineering record)	N/A	Bored piles up to approximately 150m; Driven tubular raking piles up to approximately 125m (among the deepest bridge in the world)
Toll revenue	N/A	Collection: approximately Tk. 794.53 crore from approximately 56.54 lakh vehicles
Average daily traffic	N/A	22,000 Vehicles/ day; Per vehicle approximately Tk.25 toll per vehicle
Cost Growth	N/A	In 2007: $\cong$ 197% In 2011: $\cong$ 47%
Projected National GDP impact	N/A	+(1.0-1.23)%
Projected South-West regional GDP impact	N/A	+(2.0-2.3)%

Gantt Chart of implementation schedule of the Padma Bridge Project:



1.1.6. Lessons Learned

1.1.6.1. Insights

Key Insights Area	Simple Description	Management Lesson
1. Technical & Site Risk	Soft riverbed soils and fast river currents caused major piling delays during construction.	Add 25–30% budget and schedule reserves into early project plans to handle hidden geological risks.
2. Governance Integrity	Unproven corruption claims froze donor funds and held construction work for over two years.	Keep hiring transparent to avoid procurement disputes that cause heavy delay regardless of legal outcomes.
3. Engineering Flexibility	Initial drawings could not handle changing soil conditions under 14 main river piers.	Allow project teams to adapt designs on site when ground conditions change.
4. Community Resettlement	Acquiring large land areas displaced thousands of families and threatened local livelihoods.	Provide fair compensation, relocation villages and local facilities early to build community trust and avoid project stoppages.
5. Local Skill Development	When we depend completely on outside engineers, we miss the chance to prepare our local workforce for tomorrow's large-scale projects and ongoing maintenance.	Connect local teams with foreign specialists on site, ensuring domestic engineers gain the practical experience needed for future projects.

1.1.6.2. Best Practices

Best Practices	Impact	Key action
1. Provide fair cash payments and modern relocation villages for displaced families.	Earning community trust doesn't just prevent local disputes.	Construct schools, health clinics and markets in new villages.
2. Maintain a fully backed domestic financing plan as a safety net.	Keeps construction moving without stopping if foreign loan agreements fall through.	Secure direct national budget commitments and local currency reserves early in the project.
3. Combine multiple national utility lines directly inside the main bridge structure	Saves billions in public funds by avoiding separate river crossings for gas, power and internet.	Design lower-deck utility corridors specifically for cables and pipelines .
4. Use geo-bag dumping to stabilize changing riverbanks.	Protects bridge piers and nearby villages from severe river erosion during monsoon floods.	Place durable rock layers and millions of geo-bags along 14 km of important riverbanks.

1.1.6.3. Project Management Mistakes to Avoid

Category	Mistakes	Impact	Fix
1.Engineering Planning	Finalizing bridge pier designs before finishing deep soil tests.	Forces costly redesigns in mid construction like changing 14 river piers.	Conduct thorough riverbed all testing before approving any final drawings.
2.Schedule and Cost Planning	Setting tight budgets and timelines without considering extreme weather risks.	It sets us up for unexpected costs and delayed launches which we want to avoid.	Include 25–30% time and cost buffers in the initial plan for river projects.
3. Environmental Protection	Ignoring local river wildlife and fish migration during pile driving.	Harmful noise destroys ecosystems and damages fishers' daily income.	Set up active wildlife monitoring and stop heavy work during fish breeding seasons.
4.Supply Chain and Logistics	Depending on single foreign suppliers for specialized equipment and steel parts.	Shipping delays or machine breakdowns freeze construction across the whole site.	Set up backup foreign suppliers and track custom clearances ahead of time.
5.Quality and Expert Oversight	Depending only on internal teams to solve complex deep water engineering problems.	Technical mistakes go unnoticed and risking severe structural failures.	Hire an independent panel of top international experts to review complex designs.
6.Public Communication	Keeping quiet during political debate, funding issues or project delays.	Destroys public trust and spreads false rumors.	Maintain the updates for the public and media throughout the project.

1.1.7. Conclusion :

The Padma Multipurpose Bridge is a great success for Bangladesh. The project faced many problems, like financial issues, difficult soil, river erosion, extra cost, project delays and the COVID-19 pandemic situation. However, the government fixed these problems by using their own money, better engineering solutions, strong care, river protection, and helped the affected people. The bridge was successfully completed and opened in June 2022. Now travel is much easier, communication with the south-western region is better and it helps our trade and economic growth.

1.1.7.1 Summary :

The Padma Bridge has achieved its main goal by providing a permanent road and rail connection over the Padma River. It connects Dhaka with 21 south-western districts and has greatly reduced the time needed to cross the river. This bridge also carries electricity, gas, and internet services. Although there were so many problems during construction, the project was completed successfully because of good planning, strong management, and suitable technical solutions. It is now an important part of Bangladesh's transport and economic system and it also saves our valuable time.

1.1.7.2. Future Recommendations :

- i) **Do proper testing:** Test the soil and river deeply before starting any project work. This will help make a perfect design and foundation without any mistakes.
- ii) **Keep extra time and money:** Always prepare with extra time and budget for unexpected issues like river erosion, bad weather or high cost.
- iii) **Checking regularly:** Check the bridge regularly to find the problems earlier to keep it safe for a long time.

iv) **Trying to Stop river erosion:** Work on river safety to stop the soil breakdowns and protect the bridge pillars.

v) **Help affected people:** Buy the affected people land on time who are suffering and give them fair money to the people who lose their homes to help them.

vi) **Focus on safety:** Follow strict safety rules and check the workplace regularly to keep workers safe during their work.

vii) **Train local workers:** Give better training to local engineers and workers so they can learn how to build huge projects properly.

viii) **Use new technology:** Use modern technology to keep an eye on the bridge and always prepare for repairing work.

ix) **Having a backup fund:** Always keep a backup money plan for big projects so that the work does not stop for unexpected costs.

x) **Maintain good communication :** Maintain good communication with local people to find the problems and solve this quickly.

1.1.8. References

1. The Daily Star — Economic Impact of Padma Bridge
<https://www.thedailystar.net/business/economy/news/economic-impact-padma-bridge-3092311>
2. BBC News. (2012, June 30). *World Bank cancels Bangladesh bridge loan over corruption.*
<https://www.bbc.com/news/world-south-asia-18655846>

3. Staff Correspondent. (2012, July 23). *Abul resigns, at last*. The Daily Star.
<https://www.thedailystar.net/news-detail-243336>
4. Bangladesh Bridge Authority. (n.d.). *Padma Multipurpose Bridge Project*.
<http://www.padmabridge.gov.bd/index.php>
5. Railway Technology. (n.d.). *Padma Bridge Rail Link Project, Bangladesh*.
<https://www.railway-technology.com/projects/padma-bridge-rail-link-project-bangladesh/>
6. Hossain, I. (2023, September 7). *Padma Bridge: New face of Bangladesh*. The Asian Age.
<https://dailiasianage.com/news/312081/padma-bridge-new-face-of-bangladesh>
7. The Daily Star. (2023, April 18). *Ecneec approves revision of Padma Bridge cost*.
<https://www.thedailystar.net/news/bangladesh/news/ecneec-approves-revision-padma-bridge-cost-3300081>
8. Rahman, M. T. U., & Kabir, S. B. (2022). The Padma Multipurpose Bridge: Construction challenges and sustainable management. *MIST International Journal of Science and Technology*, 10(2), 9–21. [https://doi.org/10.47981/j.mijst.10\(02\)2022.389\(09-21](https://doi.org/10.47981/j.mijst.10(02)2022.389(09-21)
9. Sultana, M. (2017, December 18). *Padma Bridge construction: Design flaws found in eight more piers*. The Financial Express.
<https://thefinancialexpress.com.bd/national/padma-bridge-construction-design-flaws-found-in-eight-more-piers-1513570464>
10. Bhaumik, S. (2013, April 18). *Bangladesh takes bold step to bridge Padma*. Al Jazeera.
<https://www.aljazeera.com/features/2013/4/18/bangladesh-takes-bold-step-to-bridge-padma>
11. Habib, W. B. (2022, June 25). *The man who steered Padma Bridge project through tough times*. The Daily Star.
<https://www.thedailystar.net/news/bangladesh/transport/news/the-man-who-steered-project-through-tough-times-3055636>
12. The Daily Star. (2022, June 24). *Padma Bridge project timeline*.
<https://www.thedailystar.net/supplements/the-daily-star-plus-padma-bridge/news/padma-bridge-project-timeline-3055826>
13. The Daily Star. (2010, January 10). *Plan to establish multi-mode communication system*.
<https://www.thedailystar.net/news-detail-120976>
14. Al Jazeera. (2012, June 30). *World Bank cancels loan for Bangladesh bridge*.
<https://www.aljazeera.com/economy/2012/6/30/world-bank-cancels-loan-for-bangladesh-bridge>
15. The Business Standard. (2020, December 10). *Padma Bridge: How the dream project came true*.
<https://www.tbsnews.net/bangladesh/infrastructure/padma-bridge-how-dream-project-came-true-169801>